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Project Proposal:
Prompt-engineering-based
Hallucination Estimation for large
language models

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Background and Motivation

In recent years, large language models (LLMs) have shown remarkable capabilities in various natural language processing (NLP) tasks, including reasoning, generation, and knowledge-intensive tasks (Zhao et al., 2025). Despite these advances, the problem of hallucinations remains one of the most significant limitations of current LLMs (Xu et al., 2025). It occurs when the model produces outputs that are reasonable and believable in form but do not coincide with the input, context, or established facts (Zhang et al., 2025). Such errors may be particularly harmful in the context of practical use, including question-answering, text summarization, and medical consultation (Huang et al., 2025). In these applications, inaccurate or fabricated information can mislead users, damage credibility, and disseminate false data, potentially becoming harmful and even dangerous (Zhang et al., 2025).

To address this issue, researchers have begun to investigate the definition and classification of hallucinations in large language models (LLMs) more systematically. For example, Zhang et al. (2025) categorize hallucinations into three categories: Input-conflicting Hallucination, Context-conflicting Hallucination, and Fact-conflicting Hallucination. They also create evaluation benchmarks to more precisely identify model errors in each domain (Zhang et al., 2025). Similarly, Huang et al. (2025) further distinguish between Faithfulness Hallucinations and Factual Hallucinations. They pointed out that these two types of hallucinations exhibit different characteristics and therefore require different mitigation strategies, particularly in retrieval-augmented generation (RAG) tasks (Huang et al., 2025). In addition, Ji et al. (2023) help differentiate between intrinsic and extrinsic hallucinations, showing that not only in LLMs but also in many natural language generation (NLG) tasks, such as summarization, generative question answering, data-to-text generation, machine translations, and even dialogue generation.

From a causal perspective, hallucinations in large language models come from several different sources. On the one hand, training data often contains noise, bias, outdated facts, or even fabricated information, which can lead models to produce incorrect or misleading outputs (Alansari & Luqman, 2025). On the other hand, decoding methods such as top-k sampling, nucleus (top-p) and temperature sampling can raise the chance of hallucina-

tions, causing the model output to deviate from the true distribution and reducing the generation quality (Kempton & Burrell, 2025). When models do not have sufficient recent or complete knowledge in fields like medicine, law, or finance, they are also more likely to give fabricated answers (Niu et al., 2024). The risks are especially high in sensitive settings. For example, in medical question-answering systems, hallucinations may cause false diagnoses or incorrect treatment suggestions (Niu et al., 2024). In addition, hallucinations weaken user trust and make it harder to use LLMs safely in real applications (Tonmoy et al., 2024). Therefore, hallucination is not only a challenge for academic research but also one of the biggest obstacles hindering the widespread deployment of large models (especially LLMs and other basic models) in real-world scenarios (Sahoo et al., 2024).

Addressing hallucinations is crucial for improving the reliability and trustworthiness of LLMs. On the one hand, reducing hallucinations can enhance the application value of LLMs in high-risk scenarios (such as healthcare, law, education, and finance) (Niu et al., 2024). On the other hand, hallucination detection and mitigation methods, including those based on uncertainty estimation (Xiong et al., 2024), retrieval enhancement (Niu et al., 2024), and prompt-engineering (Tonmoy et al., 2024), can not only improve model performance, but also provide users with a more transparent interactive experience. However, existing research is still insufficient to combine detection and mitigation, and many works focus on detecting hallucinations or proposing mitigation methods but lack a systematic closed-loop mechanism from detection to optimization (Tonmoy et al., 2024).

This project aims to build a framework based on prompt engineering for detecting, estimating and mitigating hallucinations in large language models (LLMs). The framework also strives to enhance the factual accuracy and answer reliability. This project not only brings new ideas to the research field, but also helps to improve the reliability of LLMs in practical applications.

The main reason why I chose this topic is that I have gained experience in artificial intelligence (AI) and large language models through past courses and projects. These experiences helped me have a thorough understanding of the reliability issues of large language models in practical applications, and plan to study how to improve their reliability. They also made me understand why improving the trust and safety of LLMs is

important. Through this project, I hope to obtain cutting-edge research and real-world experience, as this research direction also closely aligns with my future academic plans to pursue further studies in data science and artificial intelligence.

Aim(s) and Objectives

This project aims to detect and estimate the probability of hallucinations in model outputs and then use Prompt-engineering to effectively mitigate and correct hallucinations, thereby improving the reliability and factual consistency of large language models in practical applications.

The key objectives of this research are

- **Literature Review**

1. Systematically review the definition and classification of large language model hallucinations.
2. Analyze the main causes of hallucinations.
3. Summarize existing detection and mitigation methods, comparing their strengths and limitations.
4. Identify research gaps and propose a unique approach for this project: the lack of a closed-loop mechanism that combines detection and mitigation.

- **Datasets**

To ensure breadth and reproducibility, this project will evaluate the methods I subsequently devised on multiple tasks and datasets. Current potential datasets include:

1. **Open-domain QA:** TruthfulQA and NaturalQuestions
2. **Retrieval-augmented QA (RAG)**
3. **Summarization**

- **Models, Resources, and Risk Controls**

1. Model Selection

This project will primarily utilize open-source LLMs such as LLaMA-3 or others.

2. Resource Budget

To ensure efficiency and cost control, token and latency budgets will be set.

3. Risk Controls

In high-risk or low-confidence situations, the system will apply a default abstention or clarification strategy, refusing to answer or requesting clarification rather than generating unreliable responses.

• Detection Mechanism

1. Research and select suitable open-source hallucination detection tools or libraries for preliminary hallucination identification and verification of LLM outputs.
2. Further design and implementation of custom detection mechanisms are possible, if necessary.

• Mitigation Strategy

Based on the results of the detection phase, this project will explore and selectively adopt a range of prompt-engineering strategies to optimize model outputs. Potential approaches include, but are not limited to:

1. Prompt Rephrasing
Guide the model to give more factually consistent answers.
2. Context-Augmented Prompting
Provide more external knowledge or retrieve information to the model.
3. Chain-of-Thought Prompting
Reduce reasoning-related hallucinations.
4. One-to-Many Prompting
Generate multiple different candidate answers to the same question and select the most reliable output through consensus or majority voting.
5. Many-to-Many Prompting
Multiple related questions are cross-correlated with multiple candidate answers,

and are verified and corrected using contextual consistency and semantic alignment.

6. Innovative Strategy

Based on the above, independently design new mitigation strategies. For example, a closed-loop pipeline (estimate $\rightarrow$ mitigate $\rightarrow$ re-evaluate) is conceptually proposed and may be implemented depending on pilot study outcomes

• Detection Evaluation Process

1. Use hallucination detection methods to evaluate the model’s raw output and quantify the incidence of hallucinations.
2. Apply prompt-engineering to the same task and dataset for optimization, and again use the detection mechanism to assess the hallucination rate.
3. Compare the before-and-after detection results, using this as the core metric for measuring the effectiveness of prompt-engineering.
4. Conduct baseline experiments comparing our approach with existing non-closed-loop methods to evaluate whether the proposed closed-loop framework provides measurable improvements in hallucination detection and mitigation.
5. Conduct ablation studies by selectively removing or disabling key components to quantify the contribution of each module to overall performance.
6. Conduct significance tests to ensure that the performance improvement is meaningful.

In addition, future work may extend the evaluation to more degrees.

Work Plan

To achieve the project goals, a clear timeline and a set of specific tasks are shown in Figure 1. The preparation phase will run from late September to early November in 2025. During this time, the project will focus on writing the proposal, submitting the ethics form, and selecting large language models and datasets. A literature review will be carried out from early October to mid-December. This stage will explore how hallucinations are defined, classified, caused, and addressed in LLMs. From late October to November, the project will also begin testing detection mechanisms, including the choice of hallucination

detection tools and test datasets. An interim report will be written in December and submitted by December 16.

From January to February 2026, the work will move to building a basic closed-loop prototype. This version will include an initial detect–mitigate–reassess cycle and a first round of error analysis to check system performance. From February to March 2026, the focus will shift to developing and testing prompt-engineering strategies. Standard and creative methods will be explored to reduce hallucinations during model inference. From March to April 2026, the project will move into the evaluation phase, including performance testing, ablation studies, and benchmarking on all selected tasks and datasets. The final dissertation will be completed and refined by April 2026. After that, a demo video will be produced and preparation will begin for the Q&A session planned for late April to May 2026.

This schedule helps keep steady progress through all stages of the project. It also leaves enough time for iteration, validation, and documentation of the closed-loop hallucination mitigation framework.

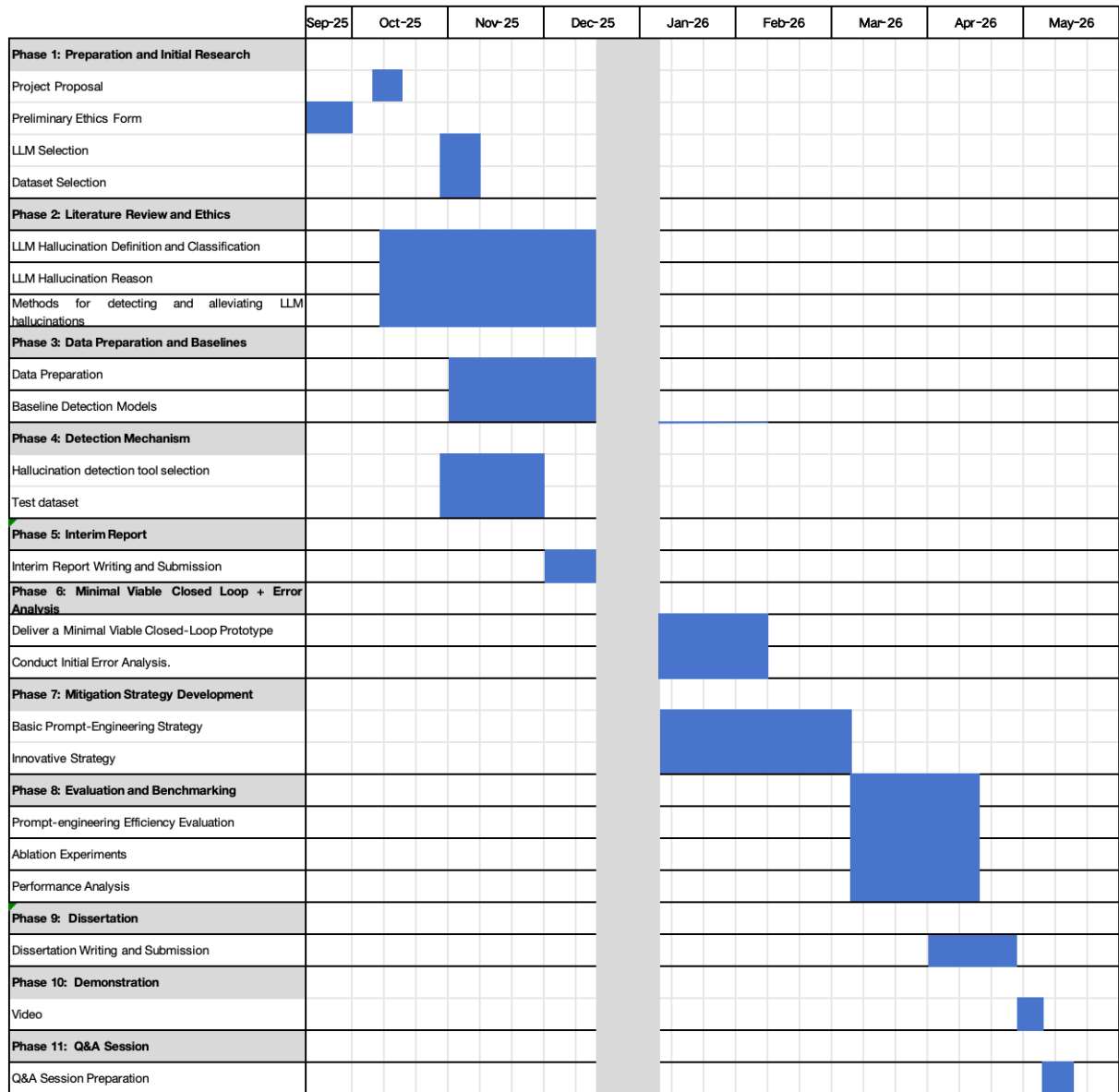


Figure 1: *Project Milestones and Time Plan*

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